Vinh Le

(650) 556-3917 · leduyvinh91@gmail.com · Santa Clara, CA · vinhduyle · VinhDuyLe

Experience

Quanta Manufacturing

Software Engineer - Fremont, CA *Feb 2026 - Present*

- Integrated the Test Automation AI Assistant (TAA) into the internal **Visual Studio Code** environment using **Model Context Protocol (MCP)**, delivering inline code explanations and automated test generation within active developer files to eliminate browser-based context switching. Tech stack: **Python 3, MCP, Google ADK**.
- Implemented an **MCP server** supporting capability discovery and tool invocation between the coding agent and backend. Tech stack: **Python (OOP, Typing), AsyncIO**.
- Built robust **JSON-RPC** request serialization and error-handling pipelines in the tool invocation handler to parse incoming queries, validate arguments, and safely route payloads to internal executors. Tech stack: **JSON-RPC 2.0, REST/RPC Protocols, Pytest**.
- Developed the TAA Planner agent using Google DeepMind's **OneTwo Planner** framework and **Gemini APIs**, encapsulating 20 complex subtools into a single conversational entry point, reducing prompt context token overhead by 20%, and isolating the IDE agent from backend tool schema changes. Tech stack: **OneTwo, Gemini API, Pydantic**.
- Engineered an asynchronous tool dispatch and observation-parsing engine with centralized trace logging and fallback handling across subtools, reducing failure triage and debugging cycles. Tech stack: **Python 3, AsyncIO, JSON Schema, Pytest**.

Frugal Innovation Hub

Software Developer - Santa Clara, CA *Jun 2024 - Jan 2026*

- Designed and implemented a **Spring Boot** search service using **Hibernate Search** to integrate **Hibernate ORM** with **Elasticsearch**, providing robust data persistence, full-text search, and query latencies under 50 ms.
- Developed a scalable books-catalog ingestion pipeline using **Java**, with **Kafka Producers** streaming data to log topics and consumers processing and batch-indexing messages into **Elasticsearch**; improved data integrity through verification, filtering, and accurate indexing.
- Developed a **Python**-based analytics service using **FastAPI** and **Pandas** to process and aggregate large-scale book metadata, delivering real-time insights and high concurrency through async event loops and connection pooling.

OpenKnect Inc.

Software Engineer Intern - Maryland *Mar 2023 - Sep 2023*

- Used **Resilience4j** with **Spring Boot** in the Media Setup Service to implement a circuit breaker for monitoring remote calls, along with retries and rate limiters, preventing cascading failures and increasing system availability to 99%.
- Developed and executed 100+ unit and integration tests across five microservices using **Java**, **JUnit**, and **Mockito**, achieving 90% test coverage and ensuring system reliability.

Khanh Loi JSC

Software Engineer - Vietnam *Jan 2020 - May 2022*

- Designed and developed 20+ **RESTful APIs** using **Spring Boot**, improving error management with custom exception handling, strengthening authentication with **JWT** and **Spring Security**, and providing comprehensive documentation with **Swagger** for hotel management, booking services, and discovery platforms.
- Implemented **Apache Kafka** as an intermediary communication layer between booking services and discovery platforms, using Kafka topics for asynchronous messaging, **Protobuf** for efficient data serialization, and **Spring Boot Kafka** integration to decouple systems and reduce data synchronization latency by 90%.

Education

Santa Clara University, Santa Clara, CA *Aug 2022 - Jan 2025*

Master of Science in Computer Science GPA: 3.8

University of Economics Ho Chi Minh City, Vietnam *Sep 2009 - May 2013*

Bachelor of Economics

Skills

AI & Agentic Systems: Model Context Protocol (MCP), Google ADK, OneTwo Planner, Gemini API, LLM Orchestration, Tool Calling, Pydantic

Programming: Java, Python, C++, JavaScript (ES6), SQL, AsyncIO, RESTful APIs, JSON-RPC 2.0, Microservices, OOP

Frameworks & Tools: Spring Boot, Spring MVC, Spring Security, Spring Cloud, Hibernate, FastAPI, Pytest, JUnit, Mockito, Apache Kafka, Elasticsearch, MySQL, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, AWS, GCP, Git, Linux, Maven, Postman